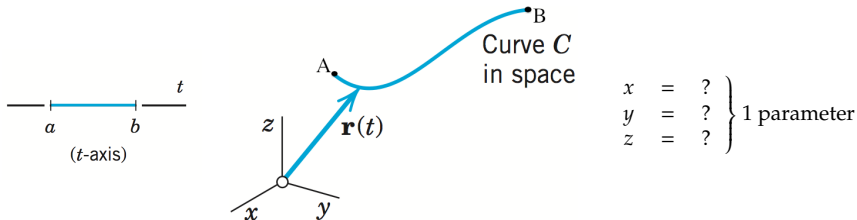


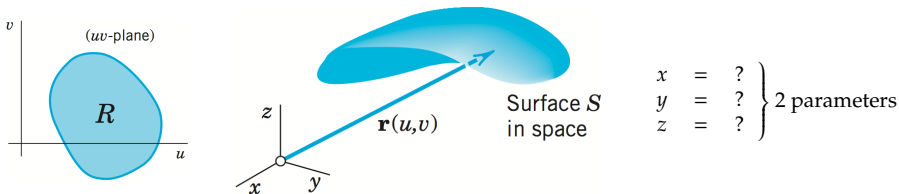
Topic: Surfaces & Surface Integrals

I. Surface Parameterization

Line Parameterization:



Surface Parameterization:



Ex: Find a parameterization of the following surfaces:

(a) $15x + 2z = 12$

$$x =$$

$$y =$$

$$z =$$

(b) $(x + 2)^2 + z^2 = 16$

$$x =$$

$$y =$$

$$z =$$

(c) $x^2 + y^2 - z^2 = 0$ in the first octant and $0 \leq z \leq 6$

$$x =$$

$$y =$$

$$z =$$

(d) $4x + 10y + 2z = 1$ where $-1 \leq y \leq 2$ and $1 \leq z \leq 5$

$$x =$$

$$y =$$

$$z =$$

(e) $-4x + 5y^2 + 7z^2 = 0$ in the first octant and $0 \leq z \leq 6$

$$x =$$

$$y =$$

$$z =$$

(f) $4x^2 + y^2 + 9z^2 = 4$

$$x =$$

$$y =$$

$$z =$$

II. Surface Tangents and Normal Vectors

Let $\mathbf{r}(u, v) = [x(u, v), y(u, v), z(u, v)]$ be a parameterization of a surface, S .

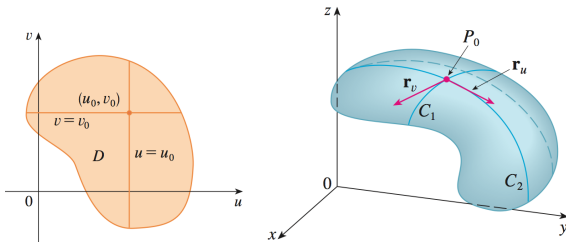
Surface Tangents:

Tangent along C_2 :

$$\mathbf{r}_u =$$

Tangent along C_1 :

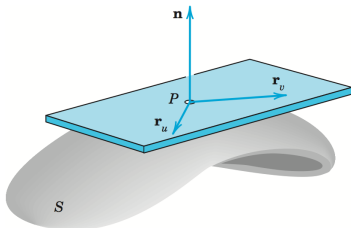
$$\mathbf{r}_v =$$



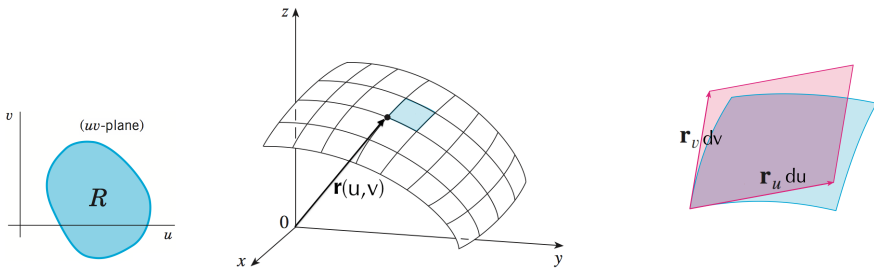
Normal:

Normal: $\mathbf{N} =$

Unit Normal: $\mathbf{n} =$



Derive the integral expression for surface area of a surface S with parameterization $\mathbf{r}(u, v)$:



Area of S =

$$\Rightarrow \text{Area}(S) =$$

$$\text{Area}(S) =$$

III. Surface Integrals

Let $\mathbf{F}$ be a vector field.

Line Integral - C is a curve

$$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} =$$

Surface Integral - S is a surface

$$\int_S \mathbf{F}(\mathbf{r}) \cdot d\mathbf{A} =$$

where $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v$ is the normal vector to the surface and $\mathbf{n} = \frac{\mathbf{N}}{|\mathbf{N}|}$ is the unit normal. R is the region in the u, v -plane (parameter plane).

IV. Flux

Flux is a very important concept when modeling physical phenomena.

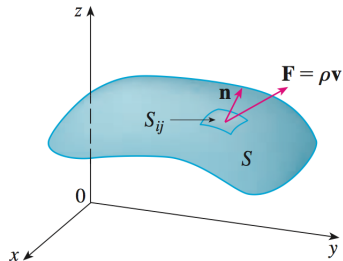
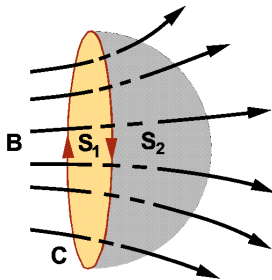
Mathematically, it is given by the surface integral

Flux through a surface $S =$

Where $\mathbf{F}$ is a vector field and $\mathbf{n}$ is the unit normal to the surface.

When dealing with flow through a surface, the vector field is typically written as:

where of the flowing substance and of the flow.



Ex: Compute the flux of water through the parabolic cylinder $S : y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 3$ if the velocity flow of water is given by $\mathbf{v} = [3z^2, 6, 6xz]$ (units: m/s). Note, the density of water is $\rho = 1 \text{ kg}/m^3$.

Set-up: we need $\mathbf{F}$, $\mathbf{r}(u, v)$, $\mathbf{r}_u$, $\mathbf{r}_v$ and $\mathbf{N}$

Since we are dealing with fluid flow, $\mathbf{F} =$

For $\mathbf{r}(u, v)$, we need

$$x =$$

$$y =$$

$$z =$$

$$\mathbf{r}(u, v) =$$

where

$$\implies \mathbf{r}_u =$$

and

$$\mathbf{r}_v =$$

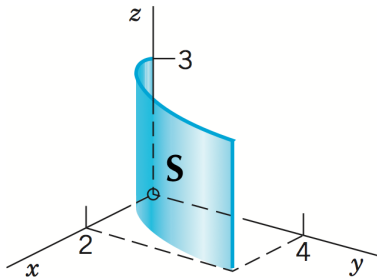
$$\mathbf{N} =$$

$$\mathbf{F}(\mathbf{r}(u, v)) =$$

$$\mathbf{F}(\mathbf{r}(u, v)) \cdot \mathbf{N} =$$

$$Flux = \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} dS$$

Translation:



Ex: Compute the surface integral of $\mathbf{F} = [e^y, -e^z, e^x]$ over the surface $S : x^2 + y^2 = 25$ for $x \geq 0$, $y \geq 0$, and $0 \leq z \leq 2$

$$x =$$

$$y =$$

$$z =$$

$$\mathbf{r}(u, v) =$$

where

$$\implies \mathbf{r}_u =$$

$$\text{and } \mathbf{r}_v =$$

$$\mathbf{N} =$$

$$\mathbf{F}(\mathbf{r}(u, v)) =$$

$$\mathbf{F}(\mathbf{r}(u, v)) \cdot \mathbf{N} =$$

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$$

V. Other Surface Integrals

Let g be a scalar-valued function.

Line Integral - C is a curve

$$\int_C g \, ds =$$

Surface Integral - S is a surface

$$\int_S g \, dA =$$

R is the region in the u, v -plane (parameter plane).

Ex: Compute the surface integral $\iint_S (1 + 9xz)^{3/2} dA$

where S is given by the parameterization: $\mathbf{r}(u, v) = [u, v, u^3]$, $0 \leq u \leq 1$, $-2 \leq v \leq 2$

$$\implies \mathbf{r}_u = \quad \text{and} \quad \mathbf{r}_v =$$

$$\mathbf{N} =$$

$$g(\mathbf{r}(u, v)) = \quad \text{and} \quad |\mathbf{N}| =$$

$$g(\mathbf{r}(u, v)) |\mathbf{N}| =$$

$$\int_S (1 + 9xz)^{3/2} dA$$